

MAHIPAL SINGH

Full Stack AI Engineer | AI Systems Builder (RAG, LLM Agents, LangChain)

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PROFESSIONAL SUMMARY

Results-driven Full Stack Engineer with 10 years of experience building scalable web applications using Angular 18+, Node.js, and MongoDB. Currently expanding expertise into AI/ML with hands-on experience in LangChain, RAG pipelines, CrewAI, OpenAI APIs, and LLM Agent development. Proven track record of delivering enterprise-grade solutions across fintech and HR-tech. Skilled in end-to-end development, cloud deployments integrating intelligent AI-powered features into modern web applications.

TECHNICAL SKILLS | CORE COMPETENCIES

Frontend	Angular 18+, AngularJS, TypeScript, JavaScript (ES6+), RxJS, HTML5, SCSS, CSS3, Bootstrap, Angular Material, PrimeNG, PWA
Backend	Node.js, Express.js, NestJS, RESTful APIs, Redis, ElectronJS, MVC Architecture
AI / LLM	LangChain, OpenAI API, CrewAI, OpenAI Agent SDK, RAG (Retrieval-Augmented Generation), Generative AI, Python, LLM Agents, Vector Databases
Databases	MongoDB, MySQL, PL/SQL
Cloud / DevOps	Docker, Nginx, Linux, Shell Scripting
Blockchain	Solidity, Web3.js, Binance Smart Chain (BSC)
Tools & VCS	Git, JIRA, Confluence, Cursor, Antigravity, VS Code, Highcharts, Google Maps API

PROFESSIONAL EXPERIENCE

Specialist Software Engineer (Full Stack) | LTM (Larsen & Toubro Infotech) Jul 2021 – Present

- Lead full-stack development using Angular 18+, Node.js, NestJS, and MongoDB for enterprise-grade platforms serving thousands of users.
- Architected and developed end-to-end features in SkillsEdge and OneView4 — HR-tech and fintech applications — with real-time data visualization using Highcharts.
- Collaborated directly with business stakeholders on requirements gathering, sprint planning, and JIRA-tracked delivery.
- Integrated AWS cloud services and implemented scalable RESTful APIs using Express and NestJS.
- Actively building AI-augmented features using LangChain, OpenAI APIs, and RAG pipelines to modernize legacy workflows.

Sr. Application Developer (Full Stack) | CompuCom System Ltd. Mar 2020 – Jul 2021

- Developed and maintained CareGuard Release 3 — a healthcare web application — using Angular, Node.js, and MongoDB.
- Built reusable Angular components, lazy-loaded modules, and optimized application performance.
- Worked in Agile/Scrum environment with cross-functional teams to deliver sprint goals on schedule.
- Implemented server-side logic and RESTful APIs with Express.js; managed MongoDB schemas and queries.

Senior UI Developer | Persistent Systems Ltd. Dec 2018 – Feb 2020

- Built dynamic single-page applications (SPAs) using AngularJS and Angular with a focus on responsiveness and UX.
- Developed Zoho CRM integrations and internal tools for the Auto Garage project.
- Worked on SAS platform as a full-stack developer using the MEAN stack.
- Improved UI performance by refactoring legacy code and adopting Angular best practices.

System Engineer | Tata Consultancy Services (TCS) Dec 2015 – Dec 2018

- Contributed to Isometrix-Ultimatix as an Oracle + Angular developer; handled complex data workflows and UI modules.

- Gained expertise in Angular, TypeScript, Oracle DB (PL/SQL), and enterprise software delivery methodologies.
- Participated in client-facing interactions, requirement analysis, and UAT support.

KEY PROJECTS

SkillsEdge — Mercer (LTM) — Full Stack Engineer

Tech Stack: Angular 18, Node.js, Express, MongoDB, Highcharts, HTML/SCSS **Duration:** Apr 2023 – Present

SkillsEdge is an integrated HR-tech platform providing consulting, technology, and skills-based talent data to enterprises.

- Built dynamic Angular 18 modules with complex form logic, lazy loading, and state management.
- Developed backend APIs using Node.js and Express; optimized MongoDB queries for performance.
- Integrated Highcharts for interactive workforce analytics dashboards.
- Triaged and resolved JIRA issues raised by stakeholders; reduced critical bug backlog by 40%.
- Exploring integration of LLM-based skill recommendation features using OpenAI and LangChain.

OneView4 — Financial Planner (LTM) — Full Stack Engineer

Tech Stack: Angular 8/9, Angular Material, NestJS, Node.js, MongoDB, AWS **Duration:** Apr 2022 – Apr 2023

OneView4 is a modern banking and retirement planning platform offering CRM-like functionality including EMI calculators, insurance management, and investment projections.

- Designed and implemented multiple Angular modules with responsive UI and role-based access control.
- Built RESTful APIs using NestJS with MongoDB integration; deployed services on AWS.
- Worked as both frontend and backend developer in a full-stack capacity within an Agile team.

Blockchain Token Development (Personal) — Blockchain Developer

Tech Stack: Solidity, Web3.js, Binance Smart Chain (BSC), JavaScript **Duration:** 2021 – 2022

Self-initiated project to design and deploy a custom cryptocurrency token on the Binance Smart Chain using Solidity smart contracts.

- Developed and deployed ERC-20-compatible smart contracts on BSC.
- Integrated Web3.js for frontend interaction with the blockchain.
- Gained deep understanding of decentralized finance (DeFi) concepts and blockchain infrastructure.

AI & EMERGING TECH SKILLS

Currently building expertise in:

- LangChain — building LLM-powered chains, agents, and tool-use pipelines
- OpenAI API & Agent SDK — integrating GPT models into full-stack applications
- CrewAI — designing multi-agent orchestration systems for autonomous task execution
- Retrieval-Augmented Generation (RAG) — implementing vector search and knowledge-augmented LLM responses
- Generative AI — practical applications in content generation, summarization, and intelligent search
- Python — scripting for AI/ML workflows, automation, and data processing

EDUCATION

Bachelor of Technology (B.Tech.) — Electronics & Communication Engineering

College of Technology, Pantnagar University | Jul 2011 – Jun 2015

ACHIEVEMENTS & INTERESTS

- National Ranker in 3x3x3 Rubik's Cube — solved in 12 seconds at national level competition.
- Ranked Top 5 nationally across 7+ puzzle categories including Pyraminx, Megaminx, 2x2–15x15 cubes.
- Solved 2x2 Rubik's Cube in 5 seconds in official competition — demonstrates analytical thinking, pattern recognition, and high focus.

PERSONAL INFORMATION

Location: Pune, India (Hometown: Dehradun, Uttarakhand) | **Nationality:** Indian | **DOB:** 08-Aug-1992